Top 54 CAT 2D & 3D LR Questions With Video Solutions

Questions

Instructions

Four cars need to travel from Akala (A) to Bakala (B). Two routes are available, one via Mamur (M) and the other via Nanur (N). The roads from A to M, and from N to B, are both short and narrow. In each case, one car takes 6 minutes to cover the distance, and each additional car increases the travel time per car by 3 minutes because of congestion. (For example, if only two cars drive from A to M, each car takes 9 minutes.) On the road from A to N, one car takes 20 minutes, and each additional car increases the travel time per car by 1 minute. On the road from M to B, one car takes 20 minutes, and each additional car increases the travel time per car by 0.9 minute.
The police department orders each car to take a particular route in such a manner that it is not possible for any car to reduce its travel time by not following the order, while the other cars are following the order.

Question 1

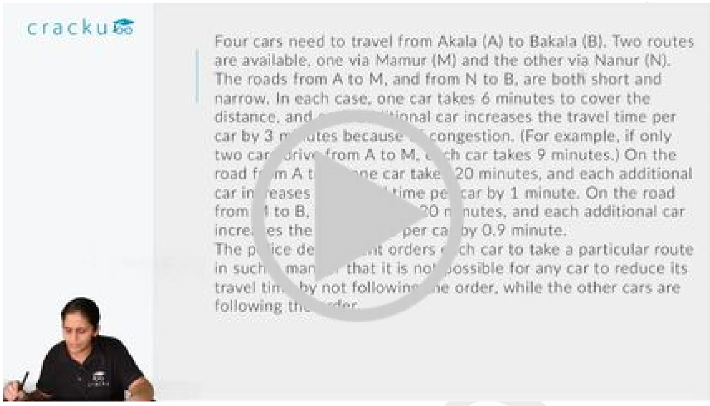
If all the cars follow the police order, what is the difference in travel time (in minutes) between a car which takes the route A-N-B and a car that takes the route A-M-B?

A 1

B 0.1

C 0.2

D 0.9



VIDEO SOLUTION

Question 2

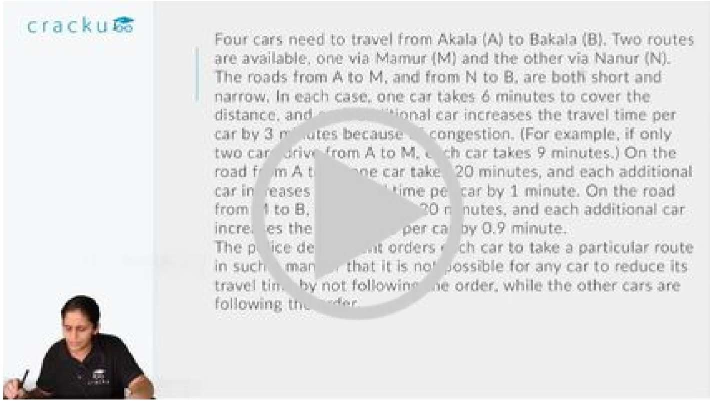
A new one-way road is built from M to N. Each car now has three possible routes to travel from A to B: A-M-B, A-N-B and A-M-N-B. On the road from M to N, one car takes 7 minutes and each additional car increases the travel time per car by j. minute. Assume that any car taking the A-M-N-B route travels the A-M portion at the same time as other cars taking the A-M-B route, and the N-B portion at the same time as other cars taking the A-N-B route.
If all the cars follow the police order, what is the minimum travel time (in minutes) from A to B? (Assume that the police department would never order all the cars to take the same route.)

A 26

B 32

C 29.9

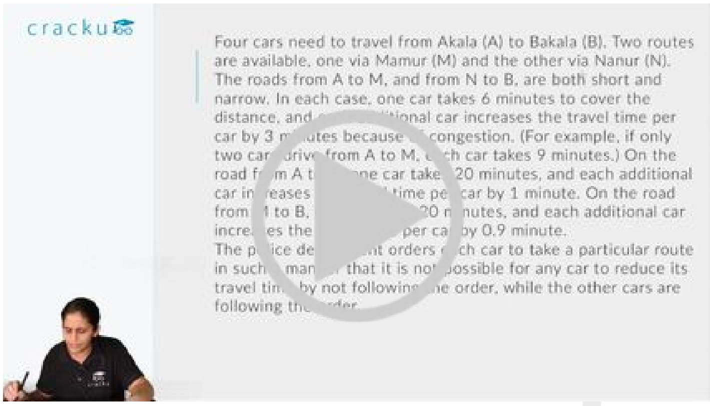
D 30



VIDEO SOLUTION

Question 3

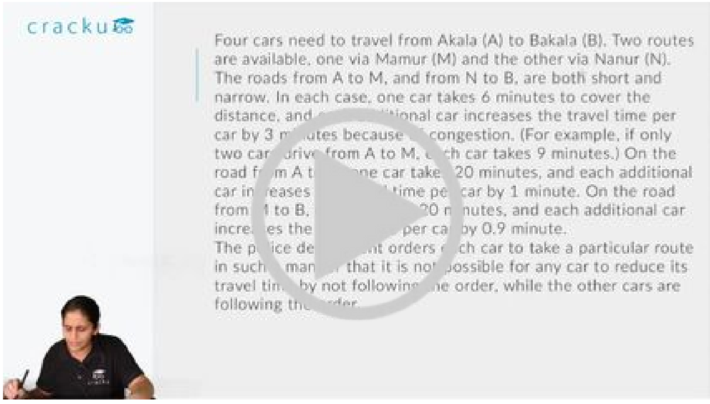
How many cars would be asked to take the route A-N-B, that is Akala-Nanur-Bakala route, by the police department?



VIDEO SOLUTION

Question 4

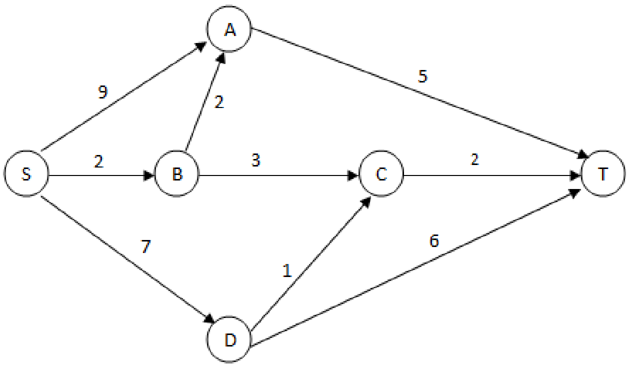
A new one-way road is built from M to N. Each car now has three possible routes to travel from A to B: A-M-B, A-N-B and A-M-N-B. On the road from M to N, one car takes 7 minutes and each additional car increases the travel time per car by 1 minute. Assume that any car taking the A-M-N-B route travels the A-M portion at the same time as other cars taking the A-M-B route, and the N-B portion at the same time as other cars taking the A-N-B route.
How many cars would the police department order to take the A-M-N-B route so that it is not possible for any car to reduce its travel time by not following the order while the other cars follow the order? (Assume that the police department would never order all the cars to take the same route.)



VIDEO SOLUTION

Instructions

A significant amount of traffic flows from point S to point T in the one-way street network shown below. Points A, B, C, and D are junctions in the network, and the arrows mark the direction of traffic flow. The fuel cost in rupees for travelling along a street is indicated by the number adjacent to the arrow representing the street. –



Motorists traveling from point S to point T would obviously take the route for which the total cost of traveling is the minimum. If two or more routes have the same least travel cost, then motorists are indifferent between them. Hence, the traffic gets evenly distributed among all the least cost routes.

The government can control the flow of traffic only by levying appropriate toll at each junction. For example, if a motorist takes the route S-A-T (using junction A alone), then the total cost of travel would be Rs 14 (i.e., Rs 9 + Rs 5) plus the toll charged at junction A.

Question 5

If the government wants to ensure that no traffic flows on the street from D to T, while equal amount of traffic flows through junctions A and C, then a feasible set of toll charged (in rupees) at junctions A, B, C,and D respectively to achieve this goal is:

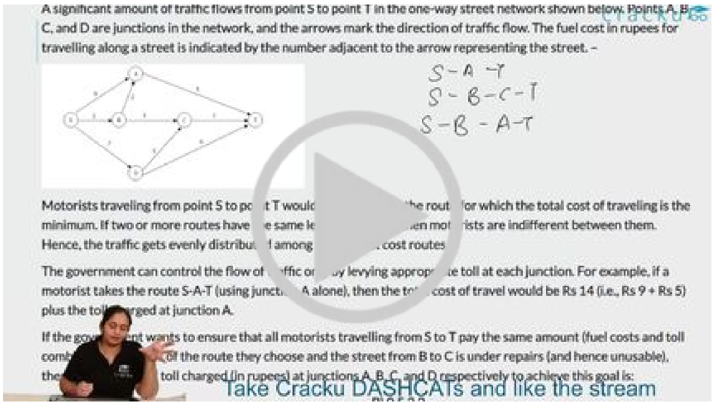
A 1,5,3,3

B 1,4,4,3

C 1,5,4,2

D 0,5,2,3

E 0,5,2,2



VIDEO SOLUTION

Question 6

If the government wants to ensure that all routes from S to T get the same amount of traffic, then a feasible set of toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

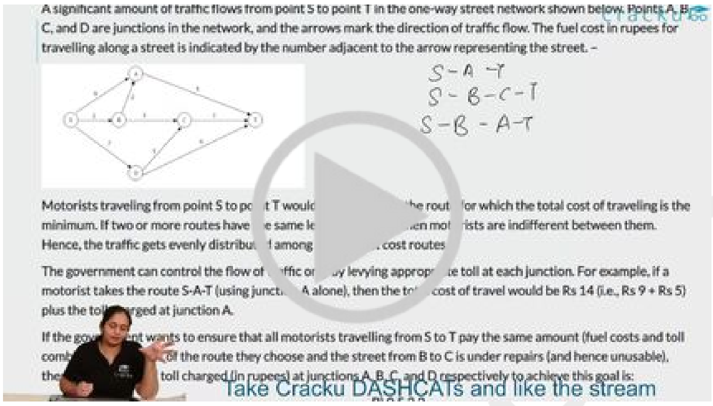
A 0, 5, 2, 2

B 0,5,4,1

C 1,5,3,3

D 1, 5, 3,2

E 1,5,4,2



VIDEO SOLUTION

Question 7

The government wants to devise a toll policy such that the total cost to the commuters per trip is minimized. The policy should also ensure that not more than 70 per cent of the total traffic passes through junction B. The cost incurred by the commuter travelling from point S to point T under this policy will be:

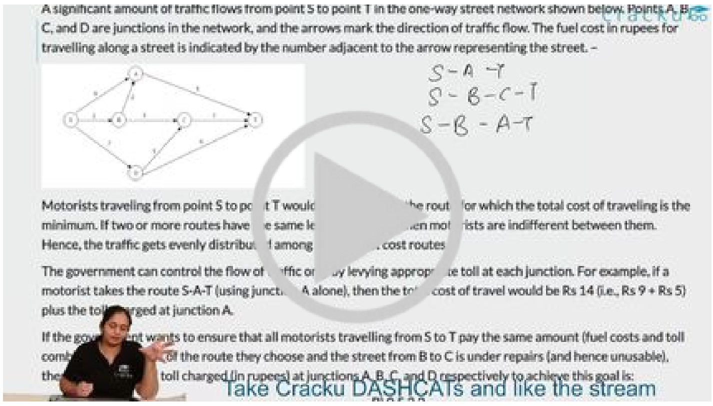
A Rs 7

B Rs 9

C Rs 10

D Rs 13

E Rs 14



VIDEO SOLUTION

Question 8

If the government wants to ensure that the traffic at S gets evenly distributed along streets from S to A, from S to B, and from S to D, then a feasible set of toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

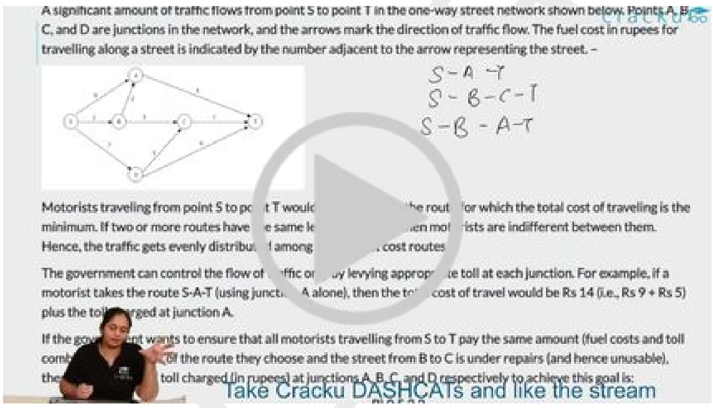
A 0,5,4,1

B 0,5,2,2

C 1,5,3,3

D 1,5,3,2

E 0,4,3,2



VIDEO SOLUTION

Question 9

If the government wants to ensure that all motorists travelling from S to T pay the same amount (fuel costs and toll combined) regardless of the route they choose and the street from B to C is under repairs (and hence unusable), then a feasible set of toll charged (in rupees) at junctions A, B, C, and D respectively to achieve this goal is:

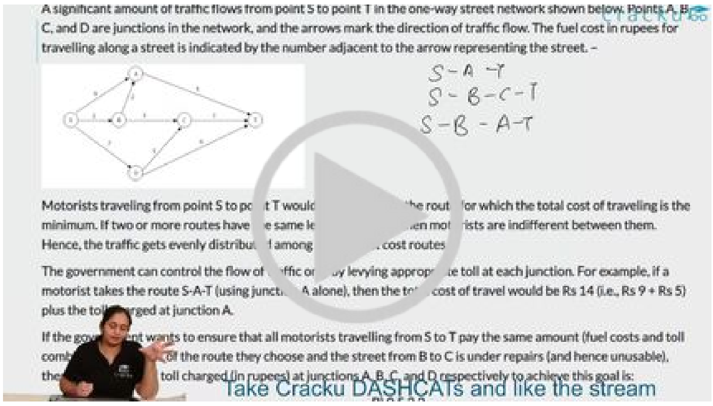
A 2,5,3,2

B 0,5,3,2

C 1,5,3,2

D 2,3,5,1

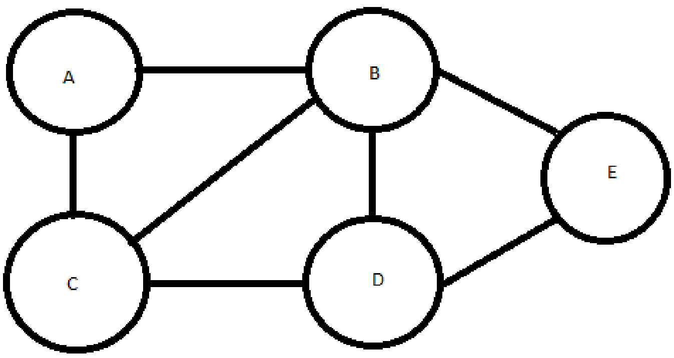
E 1,3,5,1



VIDEO SOLUTION

Instructions

There are 5 cities, A, B, C, D and E connected by 7 roads as shown in the figure below:



Design a route such that you start from any city of your choice and walk on each of the 7 roads once and only once, not necessarily returning to the city from which you started.

Question 10

For a route that satisfies the above restrictions, which of the following statements is true?

A There is no route that satisfies the above restriction.

B A route can either start at C or end at C, but not both.

C D can be only an intermediate city in the route.

D The route has to necessarily end at E.

VIEW SOLUTION

Question 11

How many different starting cities are possible such that the above restriction is satisfied?

A one

B zero

C three

D two

VIEW SOLUTION

Instructions

Direction for questions 108 and 109: Answer the questions based on the following information. In a locality, there are five small cities: A, B, C, D and E. The distances of these cities from each other are as follows. AB = 2 km AC = 2km AD > 2 km AE > 3 km BC = 2 km BD = 4 km BE = 3 km CD = 2 km CE = 3 km DE > 3 km

Question 12

If a ration shop is to be set up within 3 km of each city, how many ration shops will be required?

A 1

B 2

C 3

D 4

VIEW SOLUTION

Question 13

If a ration shop is to be set up within 2 km of each city, how many ration shops will be required?

A 2

B 3

C 4

D 5

VIEW SOLUTION

Instructions

A new airlines company is planning to start operations in a country. The company has identified ten different cities which they plan to connect through their network to start with. The flight duration between any pair of cities will be less than one hour. To start operations, the company has to decide on a daily schedule.
The underlying principle that they are working on is the following:
Any person staying in any of these 10 cities should be able to make a trip to any other city in the morning and should be able to return by the evening of the same day.

Question 14

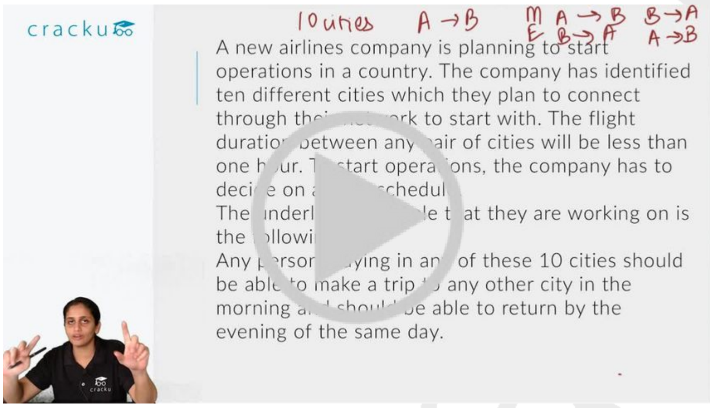
Suppose three of the ten cities are to be developed as hubs. A hub is a city which is connected with every other city by direct flights each way, both in the morning as well as in the evening. The only direct flights which will be scheduled are originating and/or terminating in one of the hubs. Then the minimum number of direct flights that need to be scheduled so that the underlying principle of the airline to serve all the ten cities is met without visiting more than one hub during one trip is:

A 54

B 120

C 96

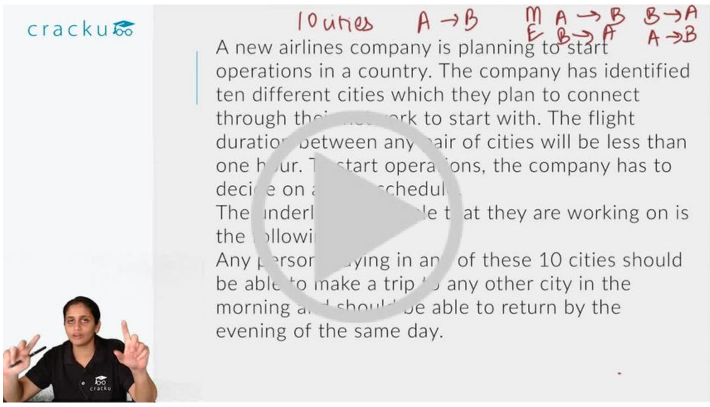
D 60



VIDEO SOLUTION

Question 15

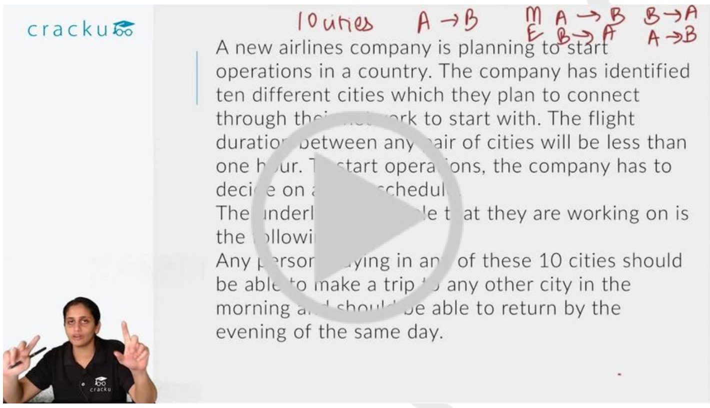
Suppose the 10 cities are divided into 4 distinct groups G1, G2, G3, G4 having 3, 3, 2 and 2 cities respectively and that G1 consists of cities named A, B and C. Further, suppose that direct flights are allowed only between two cities satisfying one of the following:
1. Both cities are in G1
2. Between A and any city in G2
3. Between B and any city in G3
4. Between C and any city in G4
Then the minimum number of direct flights that satisfies the underlying principle of the airline is:



VIDEO SOLUTION

Question 16

Suppose the 10 cities are divided into 4 distinct groups Gl, G2, G3, G4 having 3, 3, 2 and 2 cities respectively and that Gl consists of cities named A, B and C. Further, suppose that direct flights are allowed only between two cities satisfying one of the following:
1. Both cities are in G1
2. Between A and any city in G2
3. Between B and any city in G3
4. Between C and any city in G4
However, due to operational difficulties at A, it was later decided that the only flights that would operate at A would be those to and from B. Cities in G2 would have to be assigned to G3 or to G4.
What would be the maximum reduction in the number of direct flights as compared to the situation before the operational difficulties arose?



VIDEO SOLUTION

Question 17

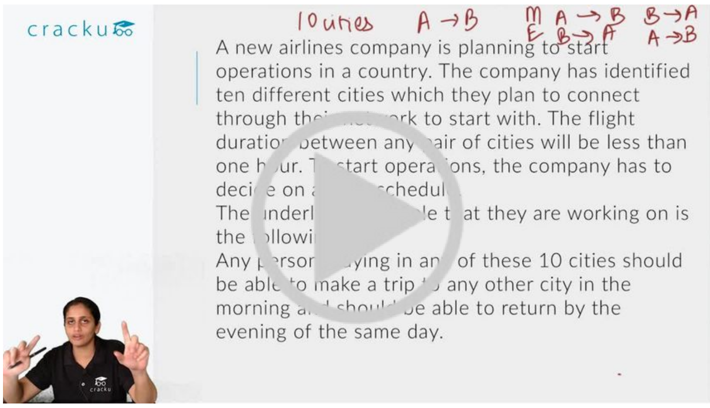
If the underlying principle is to be satisfied in such a way that the journey between any two cities can be performed using only direct (non-stop) flights, then the minimum number of direct flights to be scheduled is:

A 45

B 90

C 180

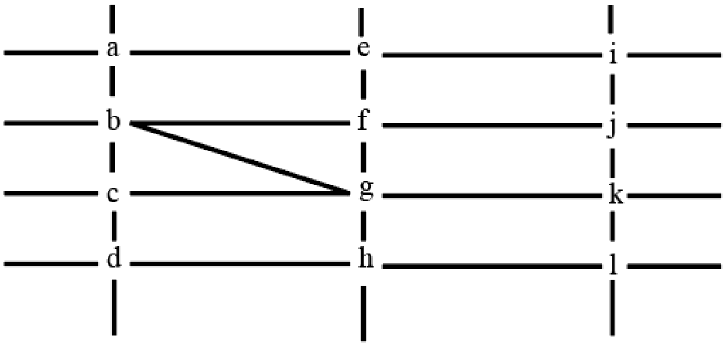
D 135



VIDEO SOLUTION

Instructions

The figure below shows the street map for a certain region with the street intersections marked from a through l. A person standing at an intersection can see along straight lines to other intersections that are in her line of sight and all other people standing at these intersections. For example, a person standing at intersection g can see all people standing at intersections b, c, e, f, h, and k. In particular, the person standing at intersection g can see the person standing at intersection e irrespective of whether there is a person standing at intersection f.



Six people U, V, W, X, Y, and Z, are standing at different intersections. No two people are standing at the same intersection.
The following additional facts are known.
1. X, U, and Z are standing at the three corners of a triangle formed by three street segments.
2. X can see only U and Z.
3. Y can see only U and W.
4. U sees V standing in the next intersection behind Z.
5. W cannot see V or Z.
6. No one among the six is standing at intersection d.

Question 18

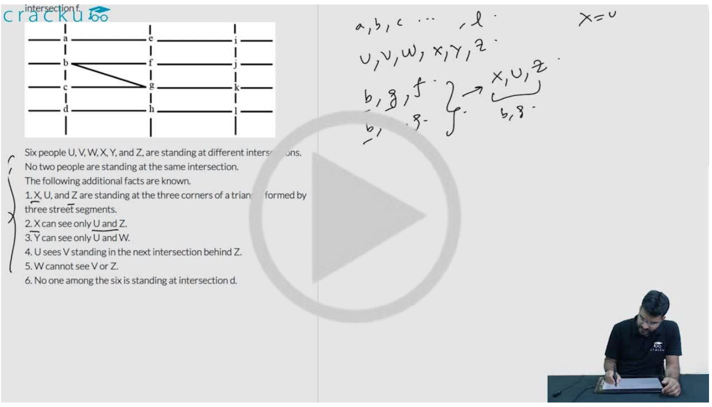
Who can V see?

A Z only

B U, W and Z only

C U and Z only

D U only



VIDEO SOLUTION

Question 19

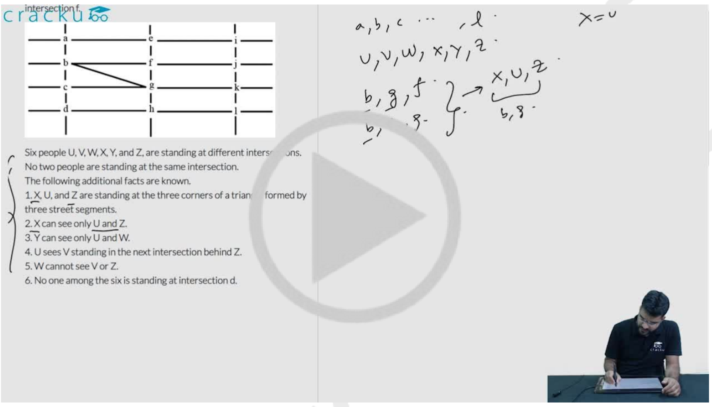
What is the minimum number of street segments that X must cross to reach Y?

A 1

B 4

C 2

D 3



VIDEO SOLUTION

Question 20

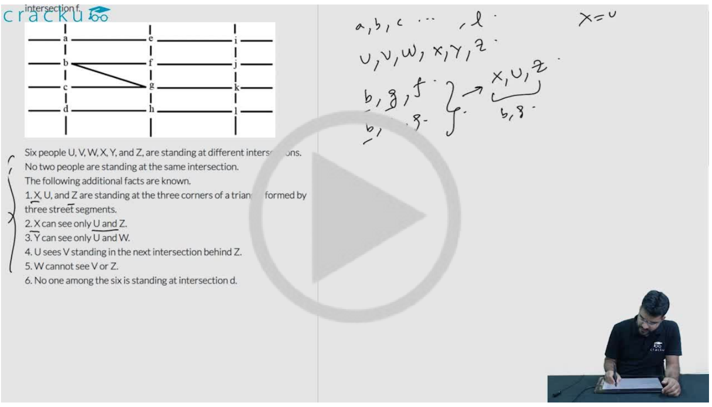
Should a new person stand at intersection d, who among the six would she see?

A W and X only

B U and W only

C U and Z only

D V and X only



VIDEO SOLUTION

Question 21

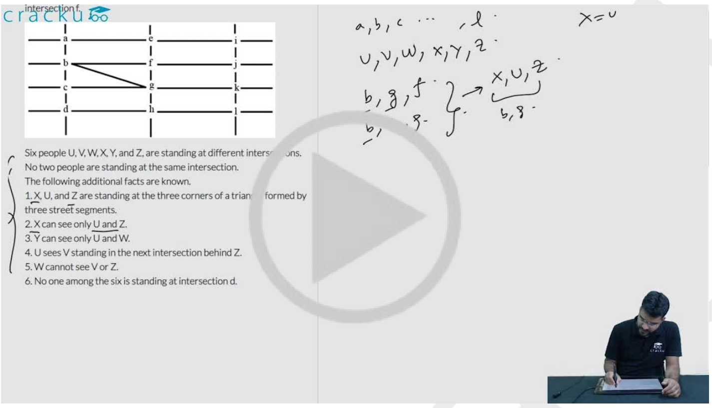
Who is standing at intersection a?

A W

B Y

C No one

D V



VIDEO SOLUTION

Instructions

In an 8 X 8 chess board a queen placed any where can attack another piece if the piece is present in the same row, or in the same column or in any diagonal position in any possible 4 directions, provided there is no other piece in between in the path from the queen to that piece.

The columns are labelled a to h (left to right) and the rows are numbered 1 to 8 (bottom to top). The position of a piece is given by the combination of column and row labels. For example, position c5 means that the piece is in c^{th} column and 5^{th} row.

Question 22

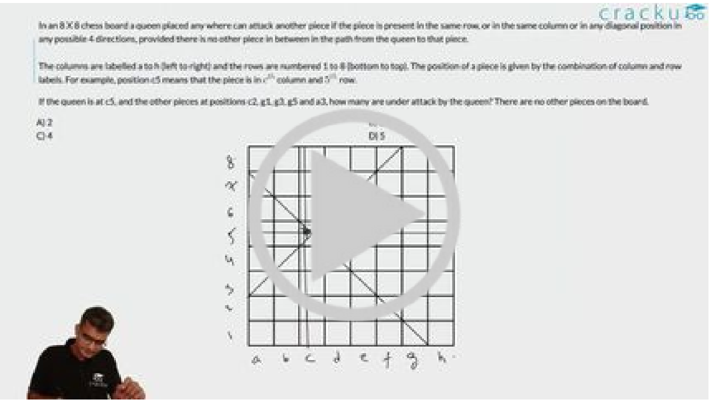
If the other pieces are only at positions a1, a3, b4, d7, h7 and h8, then which of the following positions of the queen results in the maximum number of pieces being under attack?

A f8

B a7

C c1

D d3



VIDEO SOLUTION

Question 23

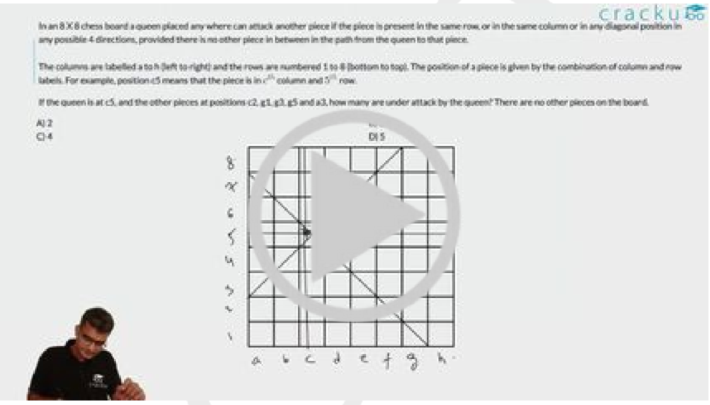
If the other pieces are only at positions a1, a3, b4, d7, h7 and h8, then from how many positions the queen cannot attack any of the pieces?

A 0

B 3

C 4

D 6



VIDEO SOLUTION

Question 24

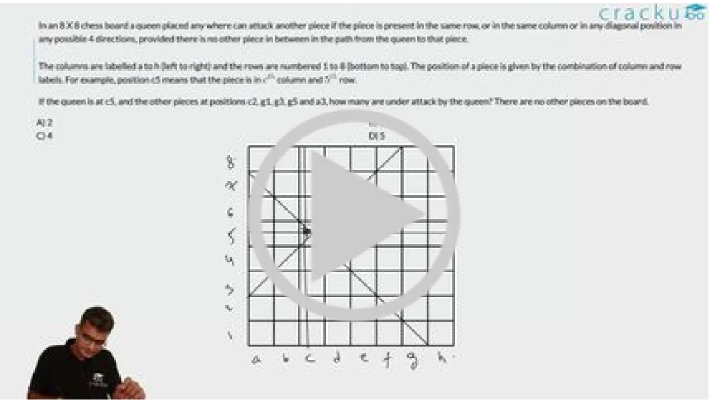
If the queen is at c5, and the other pieces at positions c2, g1, g3, g5 and a3, how many are under attack by the queen? There are no other pieces on the board.

A 2

B 3

C 4

D 5



VIDEO SOLUTION

Question 25

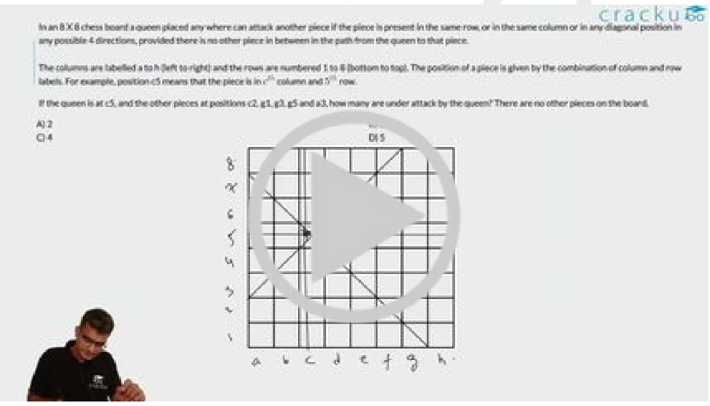
Suppose the queen is the only piece on the board and it is at position d5. In how many positions can another piece be placed on the board such that it is safe from attack from the queen?

A 32

B 35

C 36

D 37



VIDEO SOLUTION

Instructions

You are given an n×n square matrix to be filled with numerals so that no two adjacent cells have the same numeral. Two cells are called adjacent if they touch each other horizontally, vertically or diagonally. So a cell in one of the four corners has three cells adjacent to it, and a cell in the first or last row or column which is not in the corner has five cells adjacent to it. Any other cell has eight cells adjacent to it.

Question 26

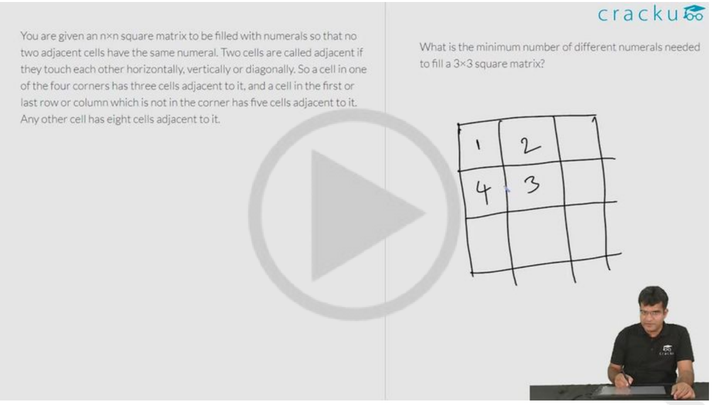
Suppose you are allowed to make one mistake, that is, one pair of adjacent cells can have the same numeral. What is the minimum number of different numerals required to fill a 5×5 matrix?

A 4

B 16

C 9

D 25



VIDEO SOLUTION

Question 27

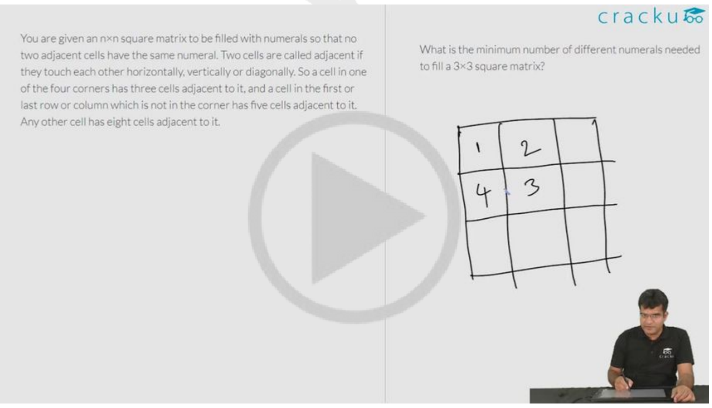
Suppose that all the cells adjacent to any particular cell must have different numerals. What is the minimum number of different numerals needed to fill a 5×5 square matrix?

A 25

B 4

C 16

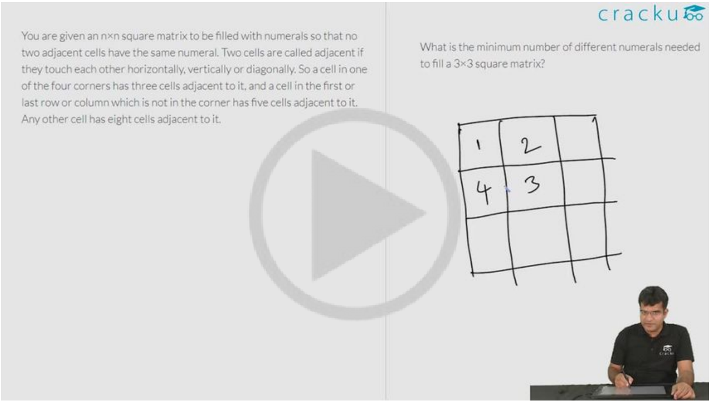
D 9



VIDEO SOLUTION

Question 28

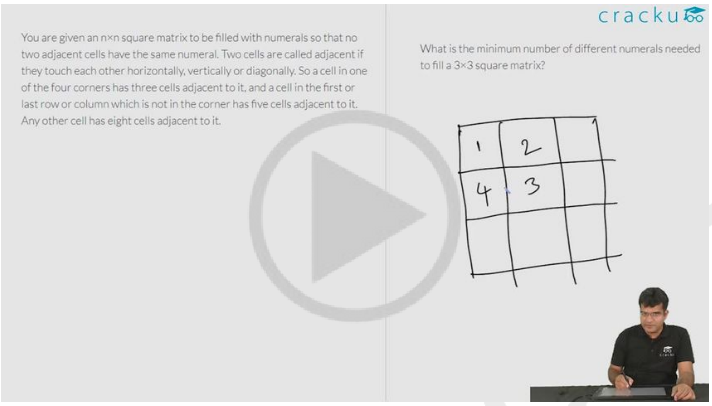
What is the minimum number of different numerals needed to fill a 5×5 square matrix?



VIDEO SOLUTION

Question 29

What is the minimum number of different numerals needed to fill a 3×3 square matrix?

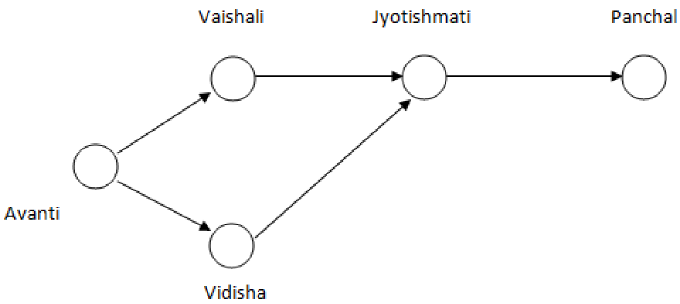


VIDEO SOLUTION

Instructions

Directions for the following three questions: Answer the questions based on the pipeline diagram below.

The following sketch shows the pipelines carrying material from one location to another. Each location has a demand for material. The demand at Vaishali is 400, at Jyotishmati is 400, at Panchal is 700, and at Vidisha is 200. Each arrow indicates the direction of material flow through the pipeline. The flow from Vaishali to Jyotishmati is 300. The quantity of material flow is such that the demands at all these locations are exactly met. The capacity of each pipeline is 1,000.



Question 30

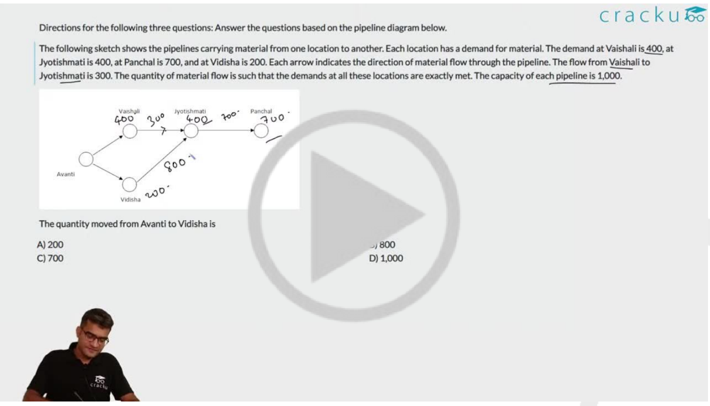
The free capacity available at the Avanti-Vaishali pipeline is

A 0

B 100

C 200

D 300



VIDEO SOLUTION

Question 31

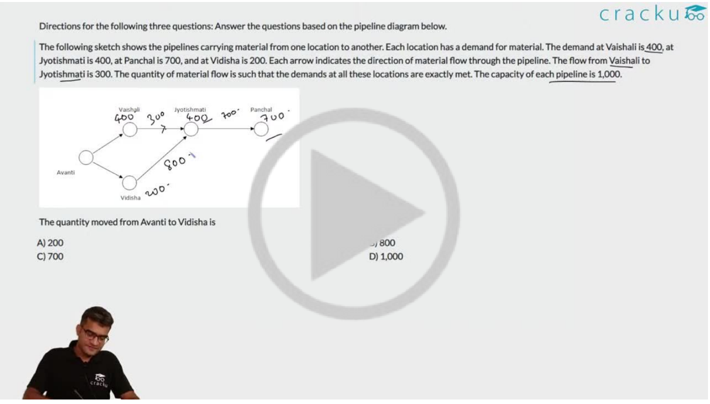
What is the free capacity available in the Avanti-Vidisha pipeline?

A 300

B 200

C 100

D 0



VIDEO SOLUTION

Question 32

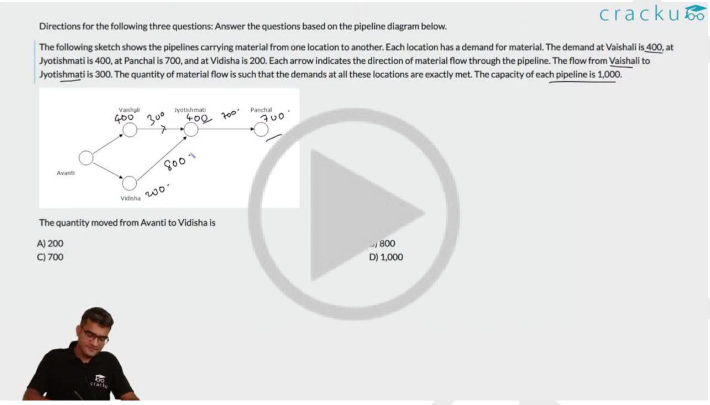
The quantity moved from Avanti to Vidisha is

A 200

B 800

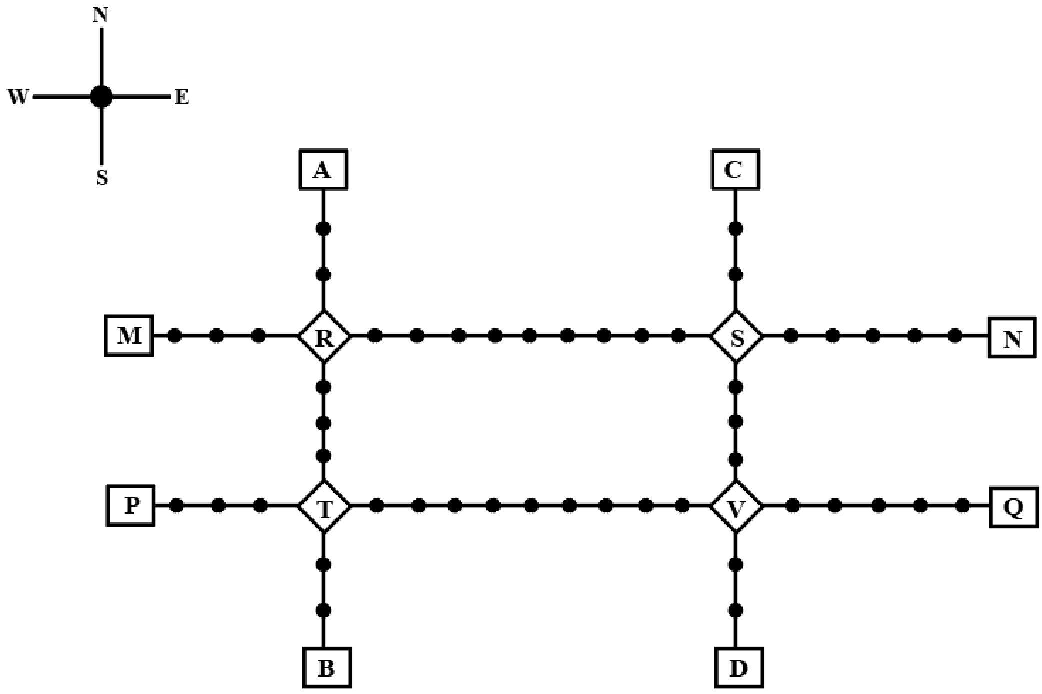
C 700

D 1,000



VIDEO SOLUTION

Instructions



Given above is the schematic map of the metro lines in a city with rectangles denoting terminal stations (e.g. A), diamonds denoting junction stations (e.g. R) and small filled-up circles denoting other stations. Each train runs either in east-west or north-south direction, but not both. All trains stop for 2 minutes at each of the junction stations on the way and for 1 minute at each of the other stations. It takes 2 minutes to reach the next station for trains going in east-west direction and 3 minutes to reach the next station for trains going in northsouth direction. From each terminal station, the first train starts at 6 am; the last trains leave the terminal stations at midnight. Otherwise, during the service hours, there are metro service every 15 minutes in the north-south lines and every 10 minutes in the east-west lines. A train must rest for at least 15 minutes after completing a trip at the terminal station, before it can undertake the next trip in the reverse direction. (All questions are related to this metro service only. Assume that if someone reaches a station exactly at the time a train is supposed to leave, (s)he can catch that train.)

Question 33

What is the minimum number of trains that are required to provide the service in this city?

VIDEO SOLUTION

Question 34

What is the minimum number of trains that are required to provide the service on the AB line (considering both north and south directions)?

VIDEO SOLUTION

Question 35

If Hari is ready to board a train at 8:05 am from station M, then when is the earliest that he can reach station N?

A 9:11 am

B 9:06 am

C 9:01 am

D 9:13 am

VIDEO SOLUTION

Question 36

If Priya is ready to board a train at 10:25 am from station T, then when is the earliest that she can reach station S?

A 11:12 am

B 11:22 am

C 11:07 am

D 11:28 am

VIDEO SOLUTION

Question 37

Haripriya is expected to reach station S late. What is the latest time by which she must be ready to board at station S if she must reach station B before 1 am via station R?

A 11:39 pm

B 11:49 am

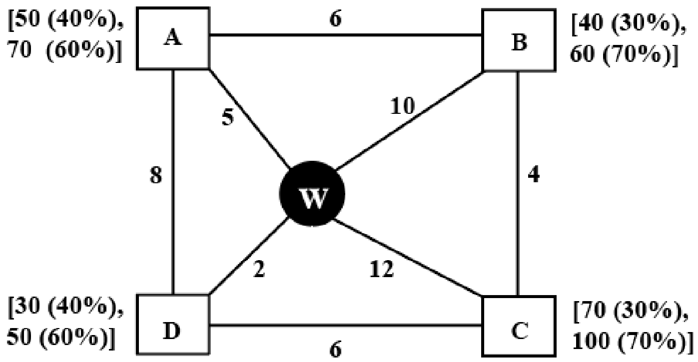
C 11:35 pm

D 11:43 pm

VIDEO SOLUTION

Instructions

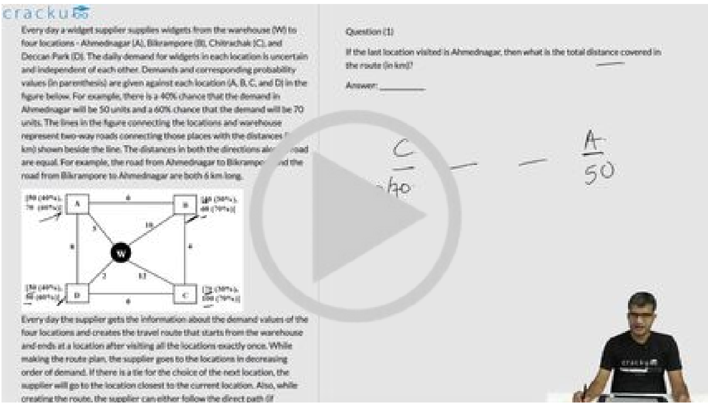
Every day a widget supplier supplies widgets from the warehouse (W) to four locations - Ahmednagar (A), Bikrampore (B), Chitrachak (C), and Deccan Park (D). The daily demand for widgets in each location is uncertain and independent of each other. Demands and corresponding probability values (in parenthesis) are given against each location (A, B, C, and D) in the figure below. For example, there is a 40% chance that the demand in Ahmednagar will be 50 units and a 60% chance that the demand will be 70 units. The lines in the figure connecting the locations and warehouse represent two-way roads connecting those places with the distances (in km) shown beside the line. The distances in both the directions along a road are equal. For example, the road from Ahmednagar to Bikrampore and the road from Bikrampore to Ahmednagar are both 6 km long.



Every day the supplier gets the information about the demand values of the four locations and creates the travel route that starts from the warehouse and ends at a location after visiting all the locations exactly once. While making the route plan, the supplier goes to the locations in decreasing order of demand. If there is a tie for the choice of the next location, the supplier will go to the location closest to the current location. Also, while creating the route, the supplier can either follow the direct path (if available) from one location to another or can take the path via the warehouse. If both paths are available (direct and via warehouse), the supplier will choose the path with minimum distance.

Question 38

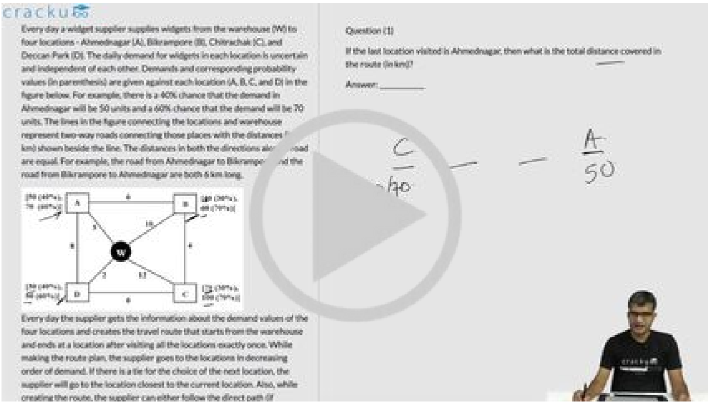
If the last location visited is Ahmednagar, then what is the total distance covered in the route (in km)?



VIDEO SOLUTION

Question 39

If the total number of widgets delivered in a day is 250 units, then what is the total distance covered in the route (in km)?



VIDEO SOLUTION

Question 40

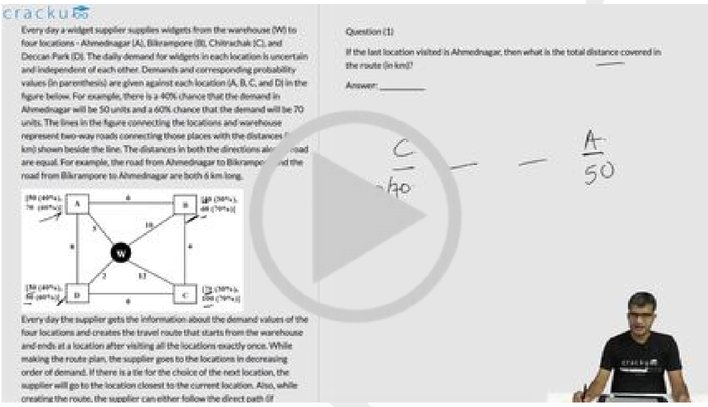
If the first location visited from the warehouse is Ahmednagar, then what is the chance that the total distance covered in the route is 40 km?

A 18%

B 5.4%

C 3.24%

D 30%



VIDEO SOLUTION

Question 41

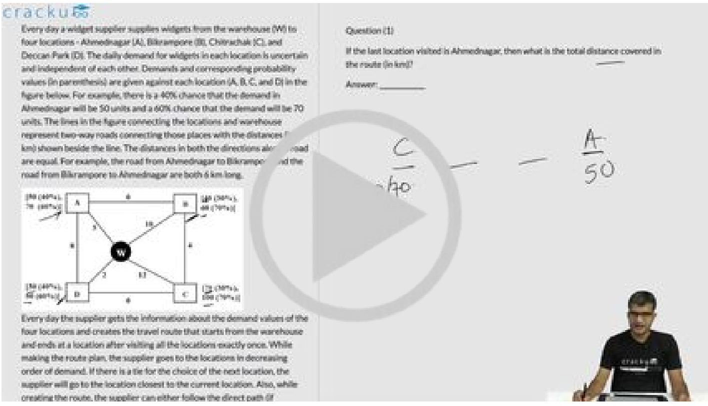
If Ahmednagar is not the first location to be visited in a route and the total route distance is 29 km, then which of the following is a possible number of widgets delivered on that day?

A 210

B 220

C 200

D 250



VIDEO SOLUTION

Question 42

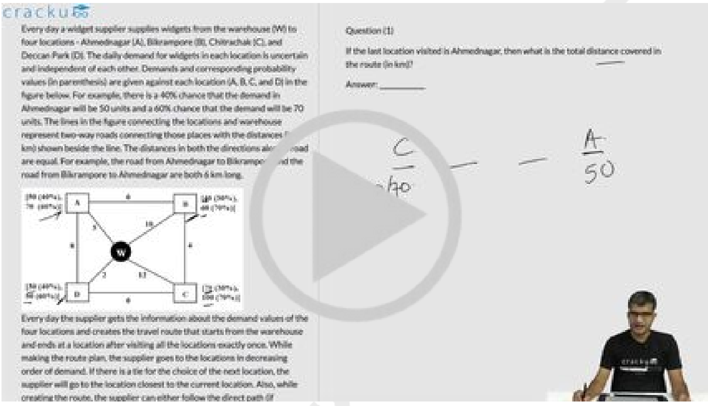
What is the chance that the total number of widgets delivered in a day is 260 units and the route ends at Bikrampore?

A 33.33%

B 10.80%

C 17.64%

D 7.56%



VIDEO SOLUTION

Instructions

A farmer had a rectangular land containing 205 trees. He distributed that land among his four daughters - Abha, Bina, Chitra and Dipti by dividing the land into twelve plots along three rows (X,Y,Z) and four Columns (1,2,3,4) as shown in the figure below:

	1	2	3	4
X	12 C			
Y	21 A			A
Z	B	C	9	28

The plots in rows X, Y, Z contained mango, teak and pine trees respectively. Each plot had trees in non-zero multiples of 3 or 4 and none of the plots had the same number of trees. Each daughter got an even number of plots. In the figure, the number mentioned in top left corner of a plot is the number of trees in that plot, while the letter in the bottom right corner is the first letter of the name of the daughter who got that plot (For example, Abha got the plot in row Y and column 1 containing 21 trees). Some information in the figure got erased, but the following is known:

1. Abha got 20 trees more than Chitra but 6 trees less than Dipti.
2. The largest number of trees in a plot was 32, but it was not with Abha.
3. The number of teak trees in Column 3 was double of that in Column 2 but was half of that in Column 4.
4. Both Abha and Bina got a higher number of plots than Dipti.
5. Only Bina, Chitra and Dipti got corner plots.
6. Dipti got two adjoining plots in the same row.
7. Bina was the only one who got a plot in each row and each column.
8. Chitra and Dipti did not get plots which were adjacent to each other (either in row / column / diagonal).
9. The number of mango trees was double the number of teak trees.

Question 43

Which column had the highest number of trees?

A 4

B 3

C Cannot be determined

D 2

VIDEO SOLUTION

Question 44

Who got the plot with the smallest number of trees and how many trees did that plot have?

A Dipti, 6 trees

B Bina, 3 trees

C Bina, 4 trees

D Abha, 4 trees

VIDEO SOLUTION

Question 45

How many pine trees did Chitra receive?

A 18

B 30

C 21

D 15

VIDEO SOLUTION

Question 46

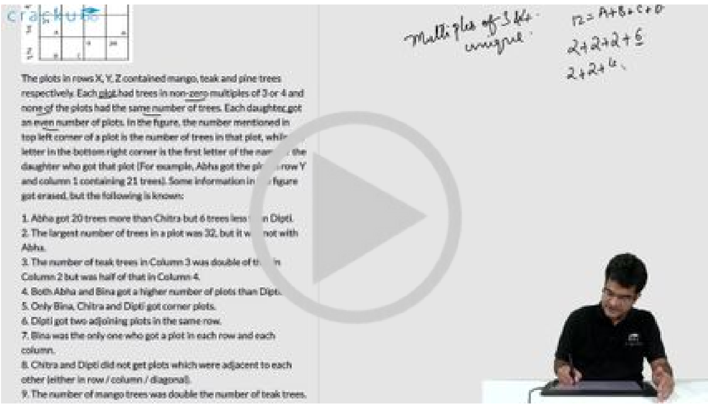
Which of the following is the correct sequence of trees received by Abha, Bina, Chitra and Dipti in that order?

A 50, 69, 30, 56

B 54, 57, 34, 60

C 44, 87, 24, 50

D 60, 39, 40, 66



VIDEO SOLUTION

Question 47

Which of the following statements is NOT true?

A Chitra got 12 mango trees

B Bina got 32 pine trees.

C Abha got 41 teak trees.

D Dipti got 56 mango trees

VIDEO SOLUTION

Question 48

How many mango trees were there in total?

A 49

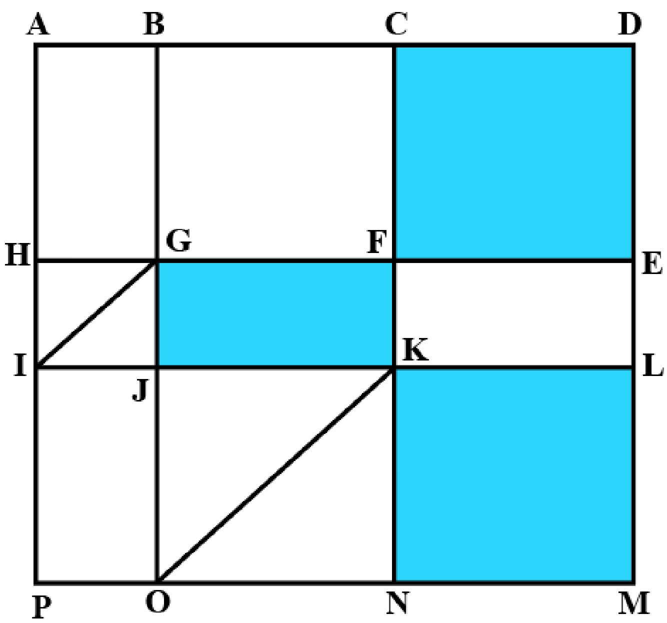
B 84

C 98

D 126

VIDEO SOLUTION

Instructions



The above is a schematic diagram of walkways (indicated by all the straight-lines) and lakes (3 of them, each in the shape of rectangles - shaded in the diagram) of a gated area. Different points on the walkway are indicated by letters (A through P) with distances being OP = 150 m, ON = MN = 300 m, ML = 400 m, EL = 200 m, DE = 400 m.

The following additional information about the facilities in the area is known.
1. The only entry/exit point is at C.
2. There are many residences within the gated area; all of them are located on the path AH and ML with four of them being at A, H, M, and L.
3. The post office is located at P and the bank is located at B.

Question 49

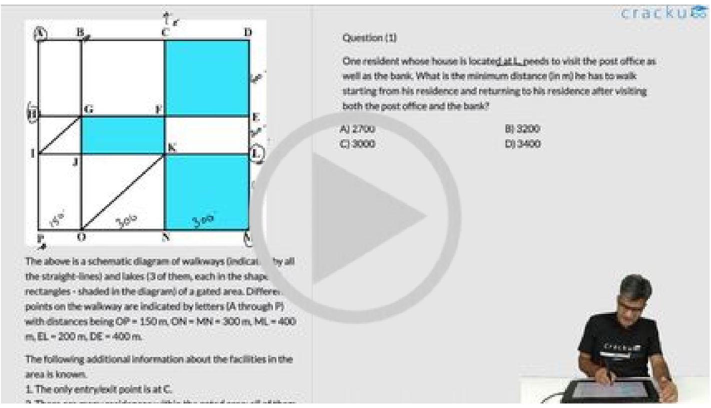
One resident whose house is located at L, needs to visit the post office as well as the bank. What is the minimum distance (in m) he has to walk starting from his residence and returning to his residence after visiting both the post office and the bank?

A 2700

B 3200

C 3000

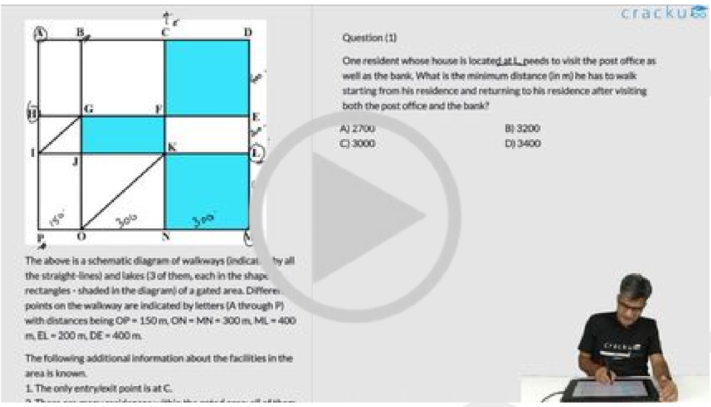
D 3400



VIDEO SOLUTION

Question 50

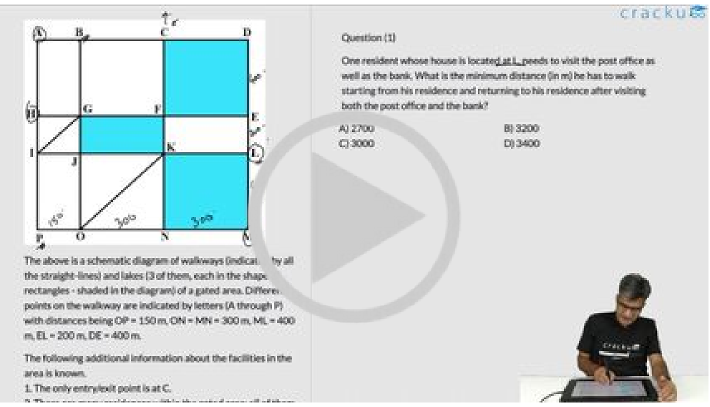
One resident takes a walk within the gated area starting from A and returning to A without going through any point (other than A) more than once. What is the maximum distance (in m) she can walk in this way?



VIDEO SOLUTION

Question 51

Visitors coming for morning walks are allowed to enter as long as they do not pass by any of the residences and do not cross any point (except C) more than once. What is the maximum distance (in m) that such a visitor can walk within the gated area?



VIDEO SOLUTION

Question 52

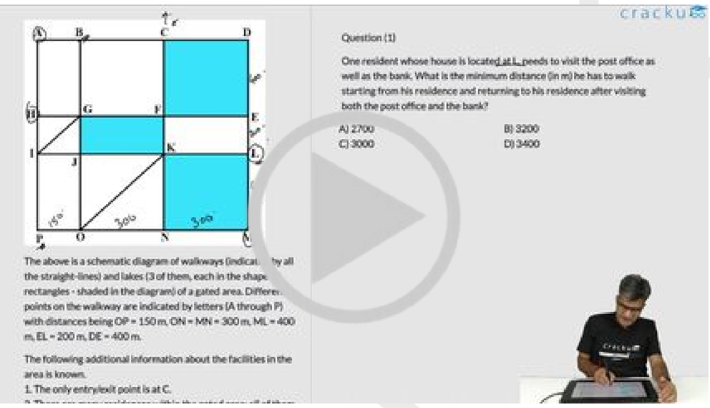
One person enters the gated area and decides to walk as much as possible before leaving the area without walking along any path more than once and always walking next to one of the lakes. Note that he may cross a point multiple times. How much distance (in m) will he walk within the gated area?

A 2800

B 3000

C 3800

D 3200



VIDEO SOLUTION

Instructions

For the following questions answer them individually

Question 53

In a six-node network, two nodes are connected to all the other nodes. Of the remaining four, each is connected to four nodes. What is the total number of links in the network?

A 13

B 15

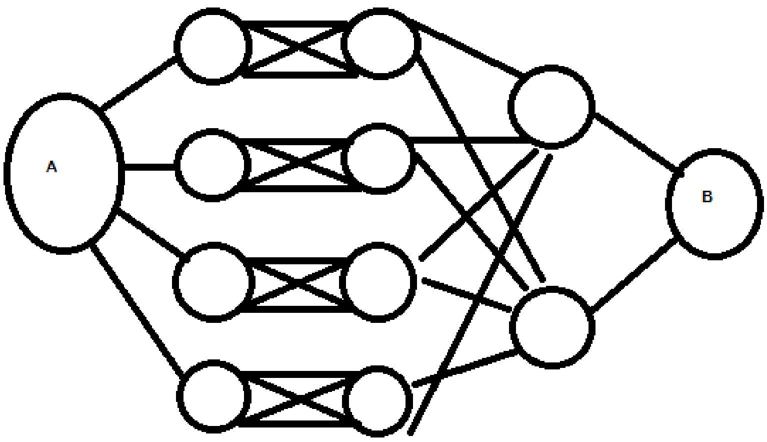
C 7

D 26

VIEW SOLUTION

Question 54

What is the total number of ways to reach A to B in the network given, such that no node is included twice and one can only move from left to right?



A 24

B 32

C 48

D 60

VIEW SOLUTION